

Transmission Spectra in EXHALE_transit.py: $H\alpha/H\beta$ from the Non-LTE $n = 2$ Population, and the Metal Resonance Doublets

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1. Overview

EXHALE_transit.py (the Transmission Probability Model shipped with EXHALE) computes transit transmission spectra by integrating the line-of-sight (LOS) optical depth on an impact-parameter grid, using Voigt profiles evaluated through the Faddeeva function, and then averaging over the projected stellar disk with optional instrument and planet-rotation convolutions. The original code handles the He I 2³S triplet (10830 Å) and the H I / D $Ly\alpha$ doublet (1215.67 Å).

This memo documents the addition of the $H\alpha$ (6562.8 Å, $n = 2 \rightarrow n = 3$) transmission spectrum. Unlike $Ly\alpha$, which absorbs out of the ground state ($n = 1$) that traces the bulk of the atomic hydrogen, $H\alpha$ absorbs out of the $n = 2$ level, whose population must be computed from a non-LTE balance. We follow Christie, Arras & Li (2013, ApJ 772, 144), including collisional excitation/de-excitation, recombination cascade, two-photon decay, ℓ -mixing, and $Ly\alpha$ radiative pumping ($1s \leftrightarrow 2p$). The $Ly\alpha$ mean intensity $J_{Ly\alpha}(r)$ is *not* produced by EXHALE; it is either read from an external text file or, by default, estimated with the simple prescription of Huang et al. (2017) — see §2.2.

The H α branch always runs; only the source of $J_{\text{Ly}\alpha}$ depends on whether `Jlya_file` is set. The existing He/Ly α behavior is unchanged.

This memo also documents the **metal resonance doublets** (Mg II h&k, Ca II H&K, Na I D; §6.), which are computed by the same LOS pipeline and produce the same three-curve spectrum figures as the H/He lines.

2. The $n = 2$ hydrogen population (Christie et al. 2013)

The $2s$ and $2p$ states are treated separately because they have very different lifetimes (the $2p \rightarrow 1s$ Ly α decay is fast, $A_{2p \rightarrow 1s} = 6.3 \times 10^8 \text{ s}^{-1}$, whereas the $2s \rightarrow 1s$ two-photon decay is slow, $A_{2s \rightarrow 1s} = 8.26 \text{ s}^{-1}$). With $n_{1s} = n_{\text{HI}}$, n_e and T taken from the EXHALE profiles, the rate-equilibrium equations are (Christie et al. Eqs. 12–13)

$$[A_{2p \rightarrow 1s} + B_{2p \rightarrow 1s} J_{\text{Ly}\alpha} + (C_{2p \rightarrow 1s} + C_{2p \rightarrow 2s})n_e + \Gamma_{2p}] n_{2p} - C_{2s \rightarrow 2p} n_e n_{2s} = (B_{1s \rightarrow 2p} J_{\text{Ly}\alpha} + C_{1s \rightarrow 2p} n_e) n_{1s} + \alpha_{2p} n_e^2, \quad (1)$$

$$[(C_{2s \rightarrow 1s} + C_{2s \rightarrow 2p})n_e + \Gamma_{2s} + A_{2s \rightarrow 1s}] n_{2s} - C_{2p \rightarrow 2s} n_e n_{2p} = C_{1s \rightarrow 2s} n_e n_{1s} + \alpha_{2s} n_e^2. \quad (2)$$

This is a 2×2 linear system in (n_{2p}, n_{2s}) ,

$$\begin{pmatrix} L_{2p} & -C_{2s \rightarrow 2p} n_e \\ -C_{2p \rightarrow 2s} n_e & L_{2s} \end{pmatrix} \begin{pmatrix} n_{2p} \\ n_{2s} \end{pmatrix} = \begin{pmatrix} S_{2p} \\ S_{2s} \end{pmatrix},$$

solved per radial cell by Cramer's rule. The total $n = 2$ density driving H α absorption is $n_2 = n_{2s} + n_{2p}$; in practice it is dominated by the metastable $2s$ state.

2.1 Rate coefficients

The forward collisional rates are taken from Christie et al. Table 2 (Janev et al. 2003); the reverse (super-elastic) rates follow by detailed balance using the statistical weights $g_{1s} = 2$, $g_{2s} = 2$, $g_{2p} = 6$. With $t_4 \equiv T/10^4 \text{ K}$:

Process	Coefficient	Expression [$\text{cm}^3 \text{ s}^{-1}$]
Case-B recomb. (Draine 2011)	α_B	$2.54 \times 10^{-13} t_4^{-0.8163-0.0208 \ln t_4}$
$\rightarrow 2s$ (incl. cascade)	α_{2s}	$(0.282 + 0.047 t_4 - 0.006 t_4^2) \alpha_B$
$\rightarrow 2p$	α_{2p}	$\alpha_B - \alpha_{2s}$
$1s \rightarrow 2s$ excitation	$C_{1s \rightarrow 2s}$	$1.21 \times 10^{-8} t_4^{-0.455} e^{-118400/T}$
$1s \rightarrow 2p$ excitation	$C_{1s \rightarrow 2p}$	$1.71 \times 10^{-8} t_4^{-0.077} e^{-118400/T}$
$2s \leftrightarrow 2p$ ℓ -mix	$C_{2s \rightarrow 2p}$	$6.21 \times 10^{-5} (\ln(T/T_0) - \gamma) / \sqrt{T}$
$2s \rightarrow 1s$ (super-elastic)	$C_{2s \rightarrow 1s}$	$1.21 \times 10^{-8} t_4^{-0.455} (g_{1s}/g_{2s})$
$2p \rightarrow 1s$ (super-elastic)	$C_{2p \rightarrow 1s}$	$1.71 \times 10^{-8} t_4^{-0.077} (g_{1s}/g_{2p})$
$2p \rightarrow 2s$ ℓ -mix	$C_{2p \rightarrow 2s}$	$C_{2s \rightarrow 2p} (g_{2s}/g_{2p})$

with $T_0 = 1.02 \text{ K}$ and $\gamma = 0.57721$ (Euler–Mascheroni). The reverse rates are written in their analytically-cancelled form: the Boltzmann factor $e^{+118400/T}$ implied by detailed balance cancels the $e^{-118400/T}$ inside $C_{1s \rightarrow 2s}/C_{1s \rightarrow 2p}$, which avoids a numerical $0 \times \infty$ at the very low temperatures that can occur at the base of the post-processed profile.

2.2 Ly α pumping and the $J_{\text{Ly}\alpha}$ input

The $1s \rightarrow 2p$ pumping rate is $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$, with the Einstein coefficient in the mean-intensity convention

$$B_{1s \rightarrow 2p} = \frac{g_{2p}}{g_{1s}} \frac{c^2}{2h\nu_{\text{Ly}\alpha}^3} A_{2p \rightarrow 1s} \simeq 8.55 \times 10^9 \text{ cgs},$$

and the stimulated-emission rate $B_{2p \rightarrow 1s} J_{\text{Ly}\alpha}$ with $B_{2p \rightarrow 1s} = (g_{1s}/g_{2p}) B_{1s \rightarrow 2p}$. Here $J_{\text{Ly}\alpha}$ is the **angle-averaged mean intensity J_ν at Ly α line center**, in cgs units $\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$, so that $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$ has units of s^{-1} .

$J_{\text{Ly}\alpha}(r)$ is obtained in one of two ways.

(1) From a file. If `Jlya_file` points to an existing two-column text file, $J_{\text{Ly}\alpha}(r)$ is read from it:

column 1	r/R_p	(same radial coordinate as the EXHALE output)
column 2	$J_{\text{Ly}\alpha}$	$[\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}]$

linearly interpolated onto the EXHALE grid and clamped to the endpoints. A template ships as `inputdata/Jlya.txt.example`.

(2) Simple estimate (default; Huang et al. 2017). If no file is given, $J_{\text{Ly}\alpha}$ is estimated following Huang, Arras, Christie & Li (2017, ApJ 851, 150), Eq. (6) and the surrounding text,

$$J_{\text{Ly}\alpha}(r) \simeq 0.1 \frac{F_{\text{LyC}}}{\Delta\nu_D(r)}, \quad \Delta\nu_D = \nu_{\text{Ly}\alpha} \frac{1}{c} \sqrt{\frac{2k_B T}{m_p}},$$

where $\Delta\nu_D$ is the local Ly α Doppler width and F_{LyC} is the *deposited* (absorbed) Lyman-continuum energy flux. The prefactor encodes the assumption that each LyC photoionization is balanced by a recombination producing one Ly α photon; Huang et al. identify F_{LyC} with the total LyC flux absorbed inside the atmosphere ($2.6 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$ for HD 189733b, page 11).

F_{LyC} from the actual stellar input. Rather than a fixed number, EXHALE_transit.py computes the absorbed LyC flux from the EXHALE inputs:

$$F_{\text{LyC}} = \underbrace{\xi \frac{10^{L_{\text{EUV}}}}{4\pi a^2}}_{\text{incident stellar LyC}} \times \underbrace{(1 - e^{-\tau_{\text{LyC}}})}_{\text{fraction absorbed}}, \quad \tau_{\text{LyC}} = \sigma_{\text{LyC}} N_{\text{HI}},$$

where L_{EUV} (the EUV-band luminosity, $E > 13.6 \text{ eV}$) and the orbital distance a are read from `input.inp`, N_{HI} is the vertical neutral-hydrogen column from the EXHALE profile, and $\sigma_{\text{LyC}} \simeq 6.3 \times 10^{-18} \text{ cm}^2$. The factor ξ is the day-night / 2D flux dilution, taken to match ATES's "2D approximate method" (`Rate/2` $\rightarrow$ 0.5, `Rate/4` $\rightarrow$ 0.25, else 1; read from `input.inp`, cf. the ξ factor of Christie+2013 / Huang+2017). For the optically-thick atomic layer $\tau_{\text{LyC}} \gg 1$, so the absorbed fraction $\rightarrow 1$ and $F_{\text{LyC}} \rightarrow \xi \times$ the incident stellar LyC flux (e.g. $\xi = 0.5$, $\simeq 5.2 \times 10^2 \text{ erg cm}^{-2} \text{ s}^{-1}$ for the HD 209458b input). A fixed value can be forced via `F_LyC_override` and the dilution via `xi_override`.

This estimate is a uniform-illumination approximation: it omits the decline of $J_{\text{Ly}\alpha}$ deep in the atmosphere captured by the full radiative-transfer treatment (Huang et al. Eq. 7) and the direct stellar Ly α component; for a self-consistent profile, supply `Jlya_file`.

2.3 $n = 2$ photoionization Γ_{2s}, Γ_{2p} (Balmer continuum)

The $n = 2$ levels are photoionized by the stellar Balmer continuum ($E > 3.4$ eV, $\lambda < 3647$ Å); this depletes n_2 and is included in Eqs. (12)–(13) through Γ_{2s}, Γ_{2p} . The Balmer-continuum field is outside EXHALE's XUV scope, so it is estimated from a diluted stellar blackbody: for a stellar effective temperature T_{star} and radius R_{star} ,

$$\Gamma_2 = \int_{\nu_2}^{\infty} \frac{F_{\nu}}{h\nu} \sigma_2(\nu) d\nu, \quad F_{\nu} = \pi B_{\nu}(T_{\star}) \left(\frac{R_{\star}}{a}\right)^2, \quad \sigma_2(\nu) = \sigma_{2,\text{th}} \left(\frac{\nu_2}{\nu}\right)^3,$$

with ν_2 the $n = 2$ threshold and $\sigma_{2,\text{th}} \simeq 1.4 \times 10^{-17} \text{ cm}^2$, and $\Gamma_{2s} \simeq \Gamma_{2p} \simeq \Gamma_2$. This reproduces Huang et al.'s Table 2 values ($\sim 20\text{--}26 \text{ s}^{-1}$) for a K dwarf, and is substantially larger for a hotter G dwarf (e.g. $\sim 140 \text{ s}^{-1}$ for HD 209458's $T_{\star} \simeq 6065 \text{ K}$), which suppresses n_2 and hence the Balmer-line depth. Setting $T_{\text{star}} \leq 0$ falls back to the manual constants Γ_{2s}/Γ_{2p} (default 0).

3. H α and H β optical depth and transmission

Once $n_2(r)$ is known, the H α transmission is computed with the same LOS machinery as Ly α . Along a chord at impact parameter b ,

$$\tau_{\nu}(b) = \int n_2(r) \sigma_{\text{H}\alpha}(\nu, r) d\ell, \quad \sigma_{\text{H}\alpha} = \frac{\sqrt{\pi} e^2}{4\pi\epsilon_0 m_e c} \frac{f_{23}}{\Delta\nu_D} H(a, x),$$

where $H(a, x) = \text{Re}[w(x + ia)]$ is the Voigt function (Faddeeva w), $\Delta\nu_D = \nu_{\text{H}\alpha} v_{\text{th}}/c$ uses the proton thermal speed, and the line-of-sight bulk velocity is folded into x exactly as for the existing lines. Atomic data: $\lambda_{\text{H}\alpha} = 6562.8$ Å (air), $f_{23} = 0.6407$, $A_{3\rightarrow 2} = 4.41 \times 10^7 \text{ s}^{-1}$ (the line is strongly Doppler-dominated, so the damping wing is negligible). The disk-averaged transmission, instrument convolution and planet-rotation convolution reuse the existing code paths.

H β ($n = 2 \rightarrow n = 4$). The same $n = 2$ population also produces H β at $\lambda_{\text{H}\beta} = 4861.35$ Å ($f_{24} = 0.119$, $A_{4\rightarrow 2} = 8.42 \times 10^6 \text{ s}^{-1}$). It is computed with an identical LOS loop and plotted/printed alongside H α . Because both lines are optically thick (saturated), the H α /H β depth ratio is much smaller than the optical-depth ratio (~ 7.3): e.g. on the converged HD 209458b run, $\text{H}\alpha \simeq 3.2\%$ and $\text{H}\beta \simeq 2.0\%$ with TPM's default star, and $\simeq 0.41\% / 0.26\%$ with HD 209458's actual $R_{\star}, T_{\star}$ (see the comparison notebook).

4. Implementation in EXHALE_transit.py

- `n2_populations(T, n1s, ne, Jlya, G2s, G2p)` solves the 2×2 system above and returns (n_{2s}, n_{2p}, n_2) in cm^{-3} . The Ly α -pump constants $B_{1s\rightarrow 2p}, B_{2p\rightarrow 1s}$ are precomputed.
- $J_{\text{Ly}\alpha}$ is read from `Jlya_file` if present, otherwise built from the Huang et al. (2017) estimate (§2.2); n_2 is then built into the symmetric (inverted+normal) radial profile `data_n2` (in m^{-3}), exactly like `data_nHI`.
- A new wavelength grid `l_nde_Ha` and an H α LOS loop fill `exp_tau_Ha`; the disk average, instrument convolution (`Instr_res_Ha`) and rotation convolution produce the plotted curves.
- The H α branch always runs (`do_Ha=True`); with the default `Jlya_file=''` it uses the Huang et al. (2017) estimate for $J_{\text{Ly}\alpha}$, with F_{LyC} taken as the absorbed stellar LyC flux (§2.2).

- **Compatibility fix.** Ion_species.txt in EXHALE carries trace-metal columns (16 columns total), so the reader was changed to usecols=range(7) to take only r and the six H/He species. This also remains correct for the old 13-column (C/O-only) and 7-column (H/He-only) outputs.

5. Validation

Run end-to-end on the converged HD 209458b EXHALE output (output/*_adv.txt), the module is stable and NaN-free and yields clean H α and H β absorption lines at 6562.8 and 4861.4 Å. With the Huang $J_{Ly\alpha}$ estimate ($\xi = 0.5$, $F_{LyC} \simeq 5.2 \times 10^2$, Γ_2 from $T_\star = 6065$ K), the depths are physically reasonable and the H α /H β ratio reflects saturation (both optically thick). The Ly α branch in the same run saturates ($\sim 100\%$ core absorption) and He I behaves as before. A worked comparison against the Jensen et al. (2012) HD 209458b observation is provided in the notebook Halpha_compare_HD209458b.ipynb.

6. Metal resonance doublets (Mg II h&k, Ca II H&K, Na I D)

The metal resonance lines are the key NUV/optical diagnostics of metal-enriched escaping atmospheres (Huang et al. 2023). They absorb out of the *ion ground state*, whose excited fine-structure levels are Boltzmann-negligible at $\sim 10^4$ K, so the lower-level density is the ion density itself ($n_{\text{lower}} \simeq n_{\text{ion}}$) — no non-LTE population calculation is needed. The ion-density profiles are read from the metal block of Ion_species_adv.txt; a metals-off run (all-zero columns) skips this section automatically.

6.1 Atomic data and machinery

Each species is treated as a **doublet**, with both components summed in a single wavelength window (Table 1; NIST f -values and A -coefficients). The optical depth uses exactly the H/He pipeline: spherical chords $\sqrt{r^2 - b^2}$ on the impact-parameter grid, LOS velocity $v_{\text{los}} = vx/r$, Voigt profiles via the Faddeeva function (vectorized over wavelength), trapezoidal τ , projected-area disk average, then the instrument convolution and the planet-rotation convolution with the depth-dependent effective radius $R_{\text{eff}} = R_p \sqrt{(h + \delta)/\delta}$ (capped at the impact-parameter boundary), $\delta = (R_p/R_\star)^2$. Figures are saved (PNG + vector PDF) via fig_name_mgii/caii/nai.

Table 1: Metal resonance doublets in EXHALE_transit.py (NIST).

species	component	λ_0 [Å]	f	A_{21} [s $^{-1}$]	window / R_{instr}
Mg II	k	2796.352	0.608	2.60×10^8	2790–2810 Å / 3×10^4
	h	2803.531	0.303	2.57×10^8	
Ca II	K	3933.663	0.6267	1.47×10^8	3927–3975 Å / Instr_res_Ha
	H	3968.469	0.3116	1.40×10^8	
Na I	D2	5889.951	0.641	6.16×10^7	5884–5902 Å / Instr_res_Ha
	D1	5895.924	0.320	6.14×10^7	

Alongside the spectra, the script prints the validated *single-component* depth table (line-center and 4 Å-band percentages for Mg II 2796, Ca II K, Na I D₂), which is the quantity used for the Huang et al. (2023) comparisons; the doublet peak differs from the single-component value only by the partner-line wing (e.g. 3.75 vs. 3.72% for Mg II on WASP-121b). Two conventions matter when comparing with Huang: their table lists the effective transit radius $R_{\text{eff}}/R_{\star} = \sqrt{(R_p/R_{\star})^2 + h}$, not a percent absorption, and Mg II is quoted in a 4 Å NUV bin while the optical lines are line-center (line-center vs. 4 Å differ by $\sim 3\times$ for the optically thick Mg II core).

For a Roche-lobe run the 3-D equipotential reconstruction (`geometry = 'triaxial'`, `roche_recon.py`) replaces the spherical chords by the full Roche geometry with per-sector LOS velocity $v_{\text{sub}}(r_{\text{eff}})x/r - \Omega y$ (wind + tidally locked rotation) and reports the same depth table.

6.2 Example spectra

Figure 1 shows the three doublet spectra for the WASP-121b worked example (`regression/tpm_wasp/`, Roche Newton solution, metals on). The optically thick cores saturate, so the two doublet members reach similar depths despite the 2:1 oscillator-strength ratio; the rotation convolution mainly shallows and broadens the narrow Na I D cores.

7. Caveats

1. **$n = 2$ photoionization.** Γ_{2s}, Γ_{2p} are now estimated from a diluted stellar-blackbody Balmer continuum (§2.3, parameter `T_star`); this is a single blackbody approximation to the true stellar near-UV spectrum. Setting `T_star` ≤ 0 reverts to the manual constants (default 0).
2. **Electron density.** $n_e = n_{\text{H II}} + n_{\text{He II}} + 2n_{\text{He III}}$ from EXHALE is used (Christie assume $n_e = n_p$); the difference is small.
3. **$J_{\text{Ly}\alpha}$ convention.** The input must be the mean intensity J_{ν} at line center in cgs (§2.2). A different convention (energy density, frequency-integrated, etc.) requires rescaling the pump constant.
4. **Wavelength/damping.** $\lambda_{\text{H}\alpha}$ is the air value 6562.8 Å; the natural width is negligible relative to the thermal Doppler width, so the line is essentially Gaussian.

Reference

- Christie, D., Arras, P., & Li, Z.-Y. 2013, ApJ, 772, 144 — “H α Absorption in Transiting Exoplanet Atmospheres” ([references/Christie_2013ApJ_772_144.pdf](#)).
- Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, ApJ, 851, 150 — “A Model of the H α and Na Transmission Spectrum of HD 189733b” ([references/Huang_2017_ApJ_851_150.pdf](#)); the $J_{\text{Ly}\alpha}$ estimate is their Eq. (6) and related text.
- Jensen, A. G., et al. 2012, ApJ, 751, 86 — transit H α spectroscopy of HD 209458b and HD 189733b (comparison target for the HD 209458b notebook).

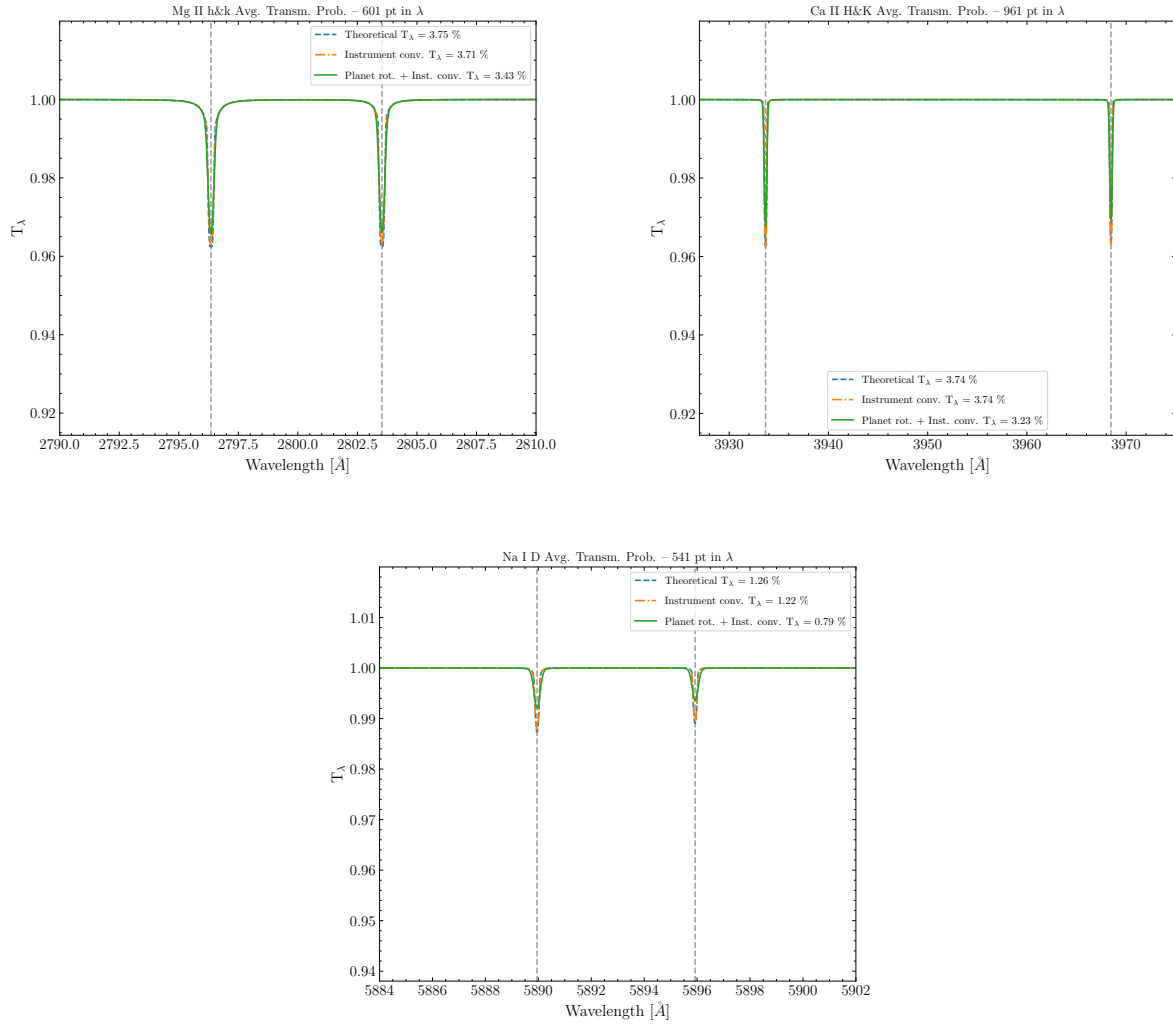


Figure 1: Metal resonance doublet transmission spectra for WASP-121b (regression/tpm_wasp/): Mg II h&k (top left), Ca II H&K (top right), and Na I D (bottom). Each panel shows the theoretical, instrument-convolved, and instrument+rotation-convolved transmission, as for the H/He lines.